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FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

1. Title of the Invention

A pipeline traversing Robot

2. Applicant(s)

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3. Preamble to the description

The following specification particularly describes the invention and the manner in which it is to be performed.

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A pipeline traversing robot

FIELD OF INVENTION

The present invention relates to the industrial gas pipe leakage detection and more particularly it relates to a pipeline traversing robot.

BACKGROUND OF INVENTION

CN117647486B discloses a combustible gas leakage concentration imaging detection device using a mid-infrared camera and thermal imaging instrument to detect flammable gas leaks, especially at elevated pipeline locations. It includes a lifting mechanism to adjust the camera height and protect the imaging components.

CN213985561U includes a special sliding device that moves along the pipeline to find and locate gas leaks using helium gas and sensors. The system works without opening or dismantling the pipeline, while our robot inspects externally. It also uses a sleeve system that fits around the pipe section to help detect leaks more precisely and safely, even in dangerous or hard-to-reach places. It can test leaks section by section, even in pipelines with bends or diameter changes.

CN117073924A is designed specifically to test the leakage rate at weld joints in spacecraft pipelines. It uses a helium mass spectrometer, a highly accurate device, to detect tiny helium gas leaks from welds. It includes a robotic arm that can move in six directions to reach complex areas on the spacecraft. Our robot is for general purpose gas leak detection, not specifically spacecraft pipelines. It focuses on automated, localized, and precise helium-based leak testing of spacecraft welds using advanced tools like laser scanners, robotic arms, and helium spectrometry.

CN118358731A discloses a submarine pipeline detection underwater robot with an ROV body, camera, lighting lamp, buffer wheels, and limiting mechanisms for crawling along underwater pipelines. However, it is designed for submarine

pipeline inspection, while our robot is an IoT-enabled gas pipe leakage detection robot for external industrial pipe inspection using magnets, wheels, gas sensors, obstacle detection, image capture, and real-time alerts.

CN114414571B uses *crawler-type rolling mechanisms* on the *sides* of the detector body, with multiple gears and toothed transmission systems. Uses air spring loop bars and hinged rods for suspension and flexibility of the crawler. Includes waterproof bearing, modular camera components, and interchangeable parts. It is powered by an internal electric motor and gear transmission system to enable the robot to move in narrow, bent, and vertical spaces without external wires.

CN114280251A focuses on finding the exact source of a gas leak using a rotating head, multi-channel gas sensors, and smart movement based on gas concentration. It also includes an air pump that draws in surrounding air, allowing the robot to detect gas from multiple directions without needing to move its entire body. The robot decides its next move based on the gas concentration gradient. It uses special gas sensor made up of many tiny sensors based on MEMS or nanomaterials, which can detect different gases more accurately.

CN116892664A uses a hybrid system combining gas and electric power to drive the robot. It has an active rotation unit that adjusts the direction and position of wheels automatically to pass complex paths. It uses spherical wheels with support-wheel mechanisms that help in turning and adapting to pipeline shape, whereas our robot uses regular wheels and castor wheels for vertical climbing and obstacle crossing. It includes a pipe diameter adjustment system allowing the robot to operate in pipes of different sizes. The robot's movement is highly dependent on video data and a central control unit making decisions based on it.

In accordance with an embodiment of the present invention, the castor wheel (9) facilitates 360-degree rotational maneuverability and assists in adapting to directional changes encountered at pipe bends.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the features of the present invention are achieved can be understood in detail, a more particular description of the invention, briefly summarized above, is provided with reference to the embodiments illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments of the invention and are therefore not to be considered limiting of its scope, as the invention may admit to other equally effective embodiments. Like reference numerals refer to like elements throughout the drawings.

Figure 1. illustrates a perspective view of a pipeline traversing robot (100) in accordance with an embodiment of the present invention, showing a three-armed symmetric base (15) configured to wrap around an exterior surface of a pipeline.

Figure 2. illustrates a top view of the pipeline traversing robot (100), showing the radial arrangement of the three arms of the base (15) and positioning of components mounted thereon.

Figure 3. illustrates a bottom view of the pipeline traversing robot (100), showing placement of a plurality of neodymium magnets (3), wheels (10), and a castor wheel (9) on the underside of the base (15).

Figure 4. illustrates a front view of the pipeline traversing robot (100), showing positioning of forward-facing components including an ultrasonic sensor (5), an imaging device (4), and a torch (14).

Figure 5. illustrates a rear view of the pipeline traversing robot (100), showing arrangement of motors (7), wheels (10), and magnetic adhesion components.

Figure 6. illustrates a left-side view of the pipeline traversing robot (100), showing spatial arrangement of structural, sensing, and actuation components.

GB2544529A primarily emphasizes pipeline inspection for structural faults, material degradation, or blockages. It uses tracked or caterpillar drives, aerofoil shapes, or differential steering to navigate. These designs are often limited to smooth surfaces or require larger body volumes and complex mechanics. It lakes IoT integration and real-time alerting for gas leakage detection.

However, there does not exist a system that provides an IoT-enabled, externally movable gas pipe leakage detection robot capable of traversing gas pipes horizontally and vertically using magnets and wheels, detecting LPG leakage in real time through onboard sensors, avoiding obstacles, capturing images in low-light conditions, storing sensed data, and sending remote alerts to a monitoring and control center.

OBJECTIVE OF THE INVENTION

- The primary objective of the present invention is to provide a pipeline traversing robot (100) configured for real-time detection and monitoring of gas leakage in industrial metallic pipelines using integrated sensing and processing mechanisms.
- Another objective of the present invention is to provide a robot having a three-armed symmetric base (15) configured to wrap around the exterior surface of a pipeline, enabling stable traversal in both horizontal and vertical orientations through magnetic adhesion and continuous radial contact of wheels (10).
- A further objective of the present invention is to provide a robot capable of overcoming surface obstacles including pipe joints and bends through a combination of ultrasonic-based obstacle detection, remote operator-assisted navigation, and a structural configuration with magnets that maintains continuous adhesion during traversal.

- Yet another objective of the present invention is to provide an adjustable mechanical configuration, wherein screws (11) enable pre-deployment adaptation of the robot to pipelines of varying diameters in the range of 150 mm to 200 mm.
- An additional objective of the present invention is to provide remote inspection capabilities through an imaging device (4) and a torch (14), enabling visual monitoring of pipeline conditions, including in low-light environments.

SUMMARY

The present invention relates to a pipeline traversing robot (100) configured for external traversal of industrial metallic gas pipelines in both horizontal and vertical orientations while overcoming surface obstacles including joints and bends. The invention may be embodied in various forms and should not be construed as limited to the embodiments set forth herein.

According to a first aspect of the present invention, the pipeline traversing robot (100) comprises a processing unit (1), a gas sensor (2), a temperature and humidity sensor (6), an ultrasonic sensor (5), an imaging device (4), a torch (14), a motor driver (8), a plurality of motors (7), wheels (10), a castor wheel (9), a plurality of neodymium magnets (3), screws (11), a base (15), a data store (13), and a monitoring and control center (12).

In accordance with an embodiment of the present invention, the base (15) comprises a three-armed symmetric structure with arms extending radially at approximately 120 degrees from a central hub, configured to wrap around the exterior surface of a pipeline such that the wheels (10) maintain simultaneous radial contact with the pipe surface from multiple directions.

In accordance with an embodiment of the present invention, the plurality of neodymium magnets (3) are mounted on the underside of the base (15) and generate magnetic adhesion force with the metallic pipe surface, thereby enabling stable

horizontal and vertical traversal while maintaining inward radial contact pressure of the wheels (10).

In accordance with an embodiment of the present invention, screws (11) provided at terminal ends of each arm enable manual angular adjustment of the arms prior to deployment, allowing the robot to adapt to pipes having diameters in the range of 150 mm to 200 mm.

In accordance with an embodiment of the present invention, the motor driver (8) controls the motors (7), each coupled to a respective wheel (10), to facilitate controlled movement of the robot along the pipeline surface.

In accordance with an embodiment of the present invention, the ultrasonic sensor (5) continuously detects obstacles and automatically halts the robot upon detecting an object within a predefined threshold distance of approximately 10 cm. The threshold distance is selected to provide sufficient stopping distance based on the response time of the drive motors and the operating speed of the robot, thereby preventing collisions during pipeline traversal.

In accordance with an embodiment of the present invention, the imaging device (4) and torch (14) enable visual inspection of the pipeline, including in low-light environments, with captured data transmitted to the monitoring and control center (12) for remote observation.

In accordance with an embodiment of the present invention, the processing unit (1) receives data from the gas sensor (2) and temperature and humidity sensor (6), processes the data using a machine learning model stored in the data store (13), determines the likelihood of gas leakage, and transmits alerts and leakage status to the monitoring and control center (12).

In accordance with an embodiment of the present invention, obstacle traversal is achieved through a combination of automated detection and manual intervention, wherein the three-armed symmetric configuration ensures that at least two arms maintain continuous adhesion and traction while the third arm negotiates the obstacle, thereby enabling stable traversal across pipe joints and bends.

Figure 7. illustrates a right-side view of the pipeline traversing robot (100), showing complementary positioning of the components relative to Figure. 6.

Figure 8. illustrates an operational view of the pipeline traversing robot (100) during vertical traversal of a pipeline, demonstrating magnetic adhesion and continuous wheel contact enabling stable climbing.

LIST OF COMPONENTS

1 - Processing Unit
2 - Gas Sensor
3 - Magnets
4 - Imaging Device
5 - Ultrasonic Sensor
6 - Temperature and Humidity Sensor
7 - Motors
8- Motor Driver
9- Castor Wheels
10 - Wheel
11- Screws
12 - Monitoring and Control Center
13 - Data Store
14- Torch
15- A Base

DETAILED DESCRIPTION OF THE DRAWINGS

Figure 1. illustrates a perspective view of the pipeline traversing robot (100). The robot (100) comprises a base (15) having a three-armed symmetric configuration, wherein each arm extends radially from a central hub at approximately 120 degrees relative to one another, enabling the robot to wrap around the exterior surface of a pipeline. A plurality of wheels (10) are mounted at terminal ends of the arms and are driven by respective motors (7) via a motor driver (8) to facilitate movement along the pipeline surface. A plurality of neodymium magnets (3) are disposed on the underside of the base (15) to provide magnetic adhesion to the metallic pipe surface, thereby enabling stable horizontal and vertical traversal. The robot (100) further comprises a processing unit (1) electrically connected to a gas sensor (2), a temperature and humidity sensor (6), and an ultrasonic sensor (5). The ultrasonic

sensor (5) is configured to detect obstacles ahead of the robot and prevent collision. An imaging device (4) and a torch (14) are mounted on the base (15) to enable visual inspection, including in low-light environments. A castor wheel (9) is provided to enable 360-degree rotational maneuverability and assist in navigating pipe bends. Sensor data is processed by the processing unit (1) and transmitted to a monitoring and control center (12) via a data store (13).

Figure 2. illustrates a top view of the pipeline traversing robot (100), showing the spatial arrangement of the three-armed symmetric base (15) and the positioning of components mounted thereon. The processing unit (1), gas sensor (2), temperature and humidity sensor (6), ultrasonic sensor (5), imaging device (4), torch (14), and motor driver (8) are positioned on the base (15) such that sensing, processing, and control functions are centrally coordinated. The wheels (10) are positioned at the terminal ends of the arms, maintaining radial alignment with the pipeline surface.

Figure 3. illustrates a bottom view of the pipeline traversing robot (100), showing the placement of the plurality of neodymium magnets (3) on the underside of each arm of the base (15), preferably proximal to the terminal ends. This configuration ensures effective magnetic adhesion and inward radial force on the wheels (10). The figure further illustrates the positioning of the castor wheel (9) and the wheels (10), which together support movement and stability during traversal.

Figure 4. illustrates a front view of the pipeline traversing robot (100), depicting the arrangement of forward-facing components including the ultrasonic sensor (5), imaging device (4), and torch (14), which collectively enable obstacle detection and visual inspection. The positioning of the gas sensor (2), temperature and humidity sensor (6), and processing unit (1) is also shown, along with the wheels (10) and castor wheel (9).

Figure 5. illustrates a rear view of the pipeline traversing robot (100), showing the placement of the motors (7), wheels (10), castor wheel (9), and neodymium magnets (3) on the underside of the base (15). The figure further depicts the arrangement of sensing and processing components mounted on the base (15).

Figure 6. illustrates a left-side view of the pipeline traversing robot (100), showing the relative positioning of the processing unit (1), gas sensor (2), ultrasonic sensor (5), imaging device (4), motor driver (8), motors (7), wheels (10), castor wheel (9), and neodymium magnets (3). This view highlights the vertical alignment of the components with respect to the pipeline surface.

Figure 7. illustrates a right-side view of the pipeline traversing robot (100), showing the placement of the temperature and humidity sensor (6) along with the processing unit (1), ultrasonic sensor (5), imaging device (4), motor driver (8), motors (7), wheels (10), castor wheel (9), and neodymium magnets (3), thereby providing a complementary view of the structural and functional arrangement.

Figure 8. illustrates the operational configuration of the pipeline traversing robot (100) during vertical traversal of a pipeline. The figure demonstrates how the three-armed symmetric base (15), in combination with the plurality of wheels (10) driven by motors (7) and the magnetic adhesion provided by neodymium magnets (3), enables the robot (100) to climb the exterior surface of the pipeline while maintaining continuous contact and stability. While traversing a bend, at least two arms maintain adhesion and traction with the pipeline surface, while the third arm adjusts its position to conform to the curvature of the pipe, thereby ensuring uninterrupted movement across the bend. The positioning of the processing unit (1) and motor driver (8) supporting this operation is also illustrated.

DETAILED DESCRIPTION OF THE INVENTION

The present invention is described hereinafter by various embodiments with reference to the accompanying drawings, wherein like reference numerals correspond to like elements throughout the description.

While the present invention is described with reference to specific embodiments and illustrative drawings, it will be understood by persons skilled in the art that the invention is not limited thereto. The drawings are not necessarily to scale, and certain components may be omitted for clarity without departing from the scope of the invention. The present invention is intended to cover all modifications, equivalents, and alternatives falling within the scope of the appended claims.

The present invention relates to a pipeline traversing robot (100) configured for external traversal of metallic pipelines for inspection and gas leakage detection, capable of operating in both horizontal and vertical orientations and overcoming surface obstacles including joints and bends.

As illustrated in the figures, the pipeline traversing robot (100) comprises a processing unit (1), a gas sensor (2), a plurality of neodymium magnets (3), an imaging device (4), an ultrasonic sensor (5), a temperature and humidity sensor (6), a plurality of motors (7), a motor driver (8), a castor wheel (9), a plurality of wheels (10), a plurality of screws (11), a monitoring and control center (12), a data store (13), and a base (15).

The base (15) comprises a three-armed symmetric structure, wherein each arm extends radially outward from a central hub at approximately 120 degrees relative to the others. The three-arm configuration represents the minimum arrangement required to establish stable tri-point contact with the pipeline surface, while the approximately 120° angular spacing maximizes radial grip symmetry and ensures uniform load distribution. This configuration enables the robot (100) to wrap around the exterior surface of a pipeline such that wheels (10) positioned at the terminal ends of each arm maintain simultaneous radial contact with the pipe surface from multiple directions.

Each arm is provided with screws (11) at its terminal end, enabling manual angular adjustment of the arms relative to the central hub prior to deployment. This allows the robot (100) to accommodate pipes of varying diameters in the range of 150 mm to 200 mm, which corresponds to standard industrial gas distribution pipelines, by forming an adjustable radial gripping configuration .

A plurality of neodymium magnets (3), selected for their high magnetic energy density (BH_max of approximately 400 kJ/m^3), are fixed on the underside of each arm, preferably proximal to the terminal ends. The magnets (3) provide sufficient magnetic adhesion to securely support the weight of the robot (100) on vertical steel pipeline surfaces while maintaining stable attachment during both horizontal and vertical traversal. The resulting magnetic force further generates inward radial

contact pressure on the wheels (10), ensuring continuous traction and preventing slippage over the pipeline surface.

Each wheel (10) is mechanically coupled to a respective motor (7), which is controlled by the motor driver (8). The coordinated actuation of the motors (7) enables controlled directional movement of the robot (100) along the pipeline surface.

The castor wheel (9), mounted on one of the arms, facilitates 360-degree rotational maneuverability and assists in adapting to directional changes, particularly while negotiating pipe bends.

The ultrasonic sensor (5) is mounted on the base (15) and continuously monitors the distance to obstacles ahead of the robot (100). Upon detecting an obstacle within a predefined threshold distance of approximately 10 cm, the ultrasonic sensor (5) transmits a signal to the processing unit (1), which in turn instructs the motor driver (8) to halt the motors (7), thereby stopping the robot (100) to prevent collision.

Subsequently, the imaging device (4) captures image data of the obstacle and transmits it via the processing unit (1) to the monitoring and control center (12). A human operator remotely evaluates the obstacle and controls the robot's movement through a mobile application interfaced with the monitoring and control center (12), enabling navigation over or around the obstacle.

The three-armed symmetric configuration ensures that during obstacle traversal, at least two arms maintain continuous adhesion and traction with the pipeline via the magnets (3) and wheels (10), while the third arm negotiates the obstacle. This sequential engagement ensures stability and uninterrupted attachment to the pipeline.

The robot (100) maintains a ground clearance of approximately 5–10 cm between the central hub and the pipeline surface. Such clearance is sufficient to accommodate standard pipe joint protrusions, typically having a height of about 3–5 cm, and other surface irregularities encountered during traversal, while ensuring continuous magnetic adhesion.

The robot (100) is equipped with a gas sensor (2) and a temperature and humidity sensor (6), both electrically connected to the processing unit (1) and configured to continuously monitor environmental and gas-related parameters in the vicinity of the pipeline.

The temperature and humidity sensor (6) continuously measures ambient temperature and humidity conditions surrounding the pipeline and transmits corresponding sensor data to the processing unit (1). Simultaneously, the gas sensor (2) continuously detects the presence and concentration of gas in the surrounding environment and transmits the sensed data to the processing unit (1).

The processing unit (1) is configured to receive data from both the gas sensor (2) and the temperature and humidity sensor (6) and to process said data using a machine learning model deployed on or accessible via the data store (13). The machine learning model analyzes multiple parameters, including gas concentration, temperature, and humidity, to determine a likelihood or probability of gas leakage. The resulting leakage status is stored in the data store (13) and transmitted to the monitoring and control center (12) for real-time monitoring and analysis. The use of multi-parameter input improves detection reliability and reduces false positives.

In addition to sensing functionality, the robot (100) comprises an imaging device (4) mounted on the base (15) and configured to capture image data of the surrounding pipeline environment. The monitoring and control center (12) communicates with the imaging device (4) via the processing unit (1) to initiate image capture. The captured image data is transmitted to and stored in the data store (13), and is subsequently retrieved by the monitoring and control center (12) for remote visual inspection, including assessment of pipeline condition and identification of potential leakage locations.

The robot (100) further comprises a torch (14) mounted on the base (15), configured to provide illumination in low-light or dark environments. The torch (14) is operable via control signals from the monitoring and control center (12) through the processing unit (1), thereby enabling effective image capture and visual inspection under insufficient lighting conditions.

The processing unit (1) is further configured to transmit operational data, including sensor readings and processed leakage status, to the monitoring and control center (12). In the event that the machine learning model determines a likelihood of gas leakage exceeding a predefined threshold of approximately 10 cm, the processing unit (1) generates and transmits an alert signal to the monitoring and control center (12), thereby enabling timely intervention.

The data store (13) is operatively coupled to the processing unit (1) and the monitoring and control center (12), and is configured to store sensor data, processed outputs, and captured image data. This arrangement enables efficient data exchange, real-time synchronization, and remote accessibility of operational information.

We claim:

1. A pipeline traversing robot (100) comprising:

a base (15) having three arms extending radially outward from a central hub, forming a three-armed symmetric structure configured at 120 degrees from one another to wrap around the exterior surface of a pipeline such that wheels (10) mounted at the terminal ends of each arm maintain simultaneous contact with the pipe surface from multiple radial directions;

a plurality of screws (11) positioned at the terminal ends of each arm of the base (15);

a plurality of neodymium magnets (3) fixed on the underside of the base (15) at a predetermined clearance from the pipe surface, configured to generate magnetic adhesion force to maintain stable contact with the exterior surface of metallic gas pipes during both horizontal and vertical traversal;

a motor driver (8) comprising of a plurality of motors (7), each electrically connected to the motor driver (8) and mechanically coupled to a respective wheel (10) being configured to provide directional movement of the robot along the pipeline;

a castor wheel (9) configured for 360-degree rotational maneuverability of the robot on the pipeline surface;

a gas sensor (2), a temperature and humidity sensor (6), an ultrasonic sensor (5) and torch (14) mounted on the base (15)

an imaging device (4) mounted on the base (15) and configured to capture image data of the surrounding pipeline environment for remote visual inspection;

a monitoring and control center (12);

a data store (13); and

a processing unit (1) electrically connected to the gas sensor (2), temperature and humidity sensor (6), ultrasonic sensor (5), imaging device (4), motor driver (8), and torch (14), and configured to receive sensor data, process the data stored in the data

store (13), and transmit leakage status and alerts to the monitoring and control center (12).

2. A pipeline traversing robot (100) as claimed in claim 1, wherein the plurality of neodymium magnets (3) are fixed on the underside of each arm of the base (15) proximal to the terminal end thereof, such that the magnetic attraction force generated between the magnets (3) and the exterior metallic surface of the gas pipe maintains inward radial contact pressure of the wheels (10) against the pipe surface , configured for adhesion of robot to the pipe along with continuous traction of the wheels (10) during traversal.

3. A pipeline traversing robot (100) as claimed in claim 1, wherein the plurality of screws (11) positioned at the terminal ends of the three arms forms a wrapping grip configuration around the exterior surface of pipes having diameters in the range of 150 mm to 200mm.

Dated this the 12th Day of July 2026

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